



Unified AI video intelligence for safer cities

Gujarat Police Innovation Challenge 2026

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Technical proposal and high-level
design

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1. Overview and model choice

Vigentra is a vendor-neutral federation layer over independent government CCTV systems. It gives a state-level operator one console across multiple department registers, Traffic Police, Municipal Corporation and any department that federates later, without moving footage between them. Each department keeps its own installation register, its own credentials and its own raw video.

THE PRINCIPLE EVERYTHING ELSE FOLLOWS FROM

Seeing that a camera exists is a federation question. Watching what it sees is the owning unit's decision. Camera metadata federates automatically on validation. Footage never federates: it is asked for, granted personally, time-boxed, and revocable.

This document describes a system that runs. Every component below is implemented, tested and demonstrable. Where a figure is an engineering estimate rather than a measurement, it is labelled as one, and section 10 separates the two deliberately. A design document that does not distinguish measured from projected is not one a deploying engineer can act on.

1.1 How this document maps to the required high-level design

Required content	Section
Overall solution architecture, diagrams and component interactions	2
Integrating heterogeneous cameras, NVRs and Video Management Systems	3
Ingesting, processing and managing live streams from dispersed sites	4
Watchlist integration and continuous correlation into real-time alerts	5

Required content	Section
AI video analytics: ANPR, FRS, object detection, person and vehicle tracking	6
Alert generation and notification workflow, prioritisation, visualisation, interaction	7
Scalability, interoperability, security and performance at 80,000 cameras	8
Technical prerequisites, assumptions and information required from departments	9

Table 1. Each required element of the high-level design, and where it is answered.

1.2 Model choice and justification

The challenge offers four reference models. Vigentra adopts two in full, one as a surface, and rejects one on arithmetic.

Model	Verdict	Reasoning
1. Registry and GIS	Adopted in full	Nothing else can be built without knowing which cameras exist, where, owned by whom, in what state of repair. It is also the only part that delivers value with no analytics at all: an accurate statewide inventory with coverage-gap analysis is useful on day one, and nobody has one today.
2. Unified viewing	Adopted as a surface	Operators need one console, and that part is right. Connecting the console directly to 26 departmental VMS platforms is not: it puts vendor-specific logic in the viewer and makes every new vendor a change to the thing operators use.

Model	Verdict	Reasoning
3. VMS federation middleware	Adopted as the integration	An adapter per vendor, normalising to one canonical model, makes vendor differences invisible above the boundary and lets a department be onboarded without touching anything downstream.
4. Central VMS	Rejected	Centralising 80,000 video streams is roughly 160 Gbps sustained and 1.7 PB per day. It also concentrates every camera credential in one place and makes ambient statewide viewing the default, which is the worst available privacy posture. Section 8.1 shows the arithmetic.

Table 2. Reference model assessment and the reasoning behind each verdict.

So: Model 1 as the foundation, Model 3 as the integration layer, Model 2's console as the surface, and analytics at the edge. Video is pulled on demand per authorised session, never streamed centrally by default. Inference goes to the video instead of video coming to the centre.

The constraint that shapes every decision is that the departments keep running their own systems. A design that requires 26 departments to migrate onto one VMS before anything works is a design that produces nothing for three years. Vigentra is built so that the first department to federate gets value on the day it does, and the twenty-sixth changes nothing about how the first works.

2. Overall solution architecture

2.1 Four tiers, one boundary that matters

80,000 cameras is not one large system. It is four tiers with very different constraints, and conflating them is what produces the answer that a lot of GPUs will be needed.

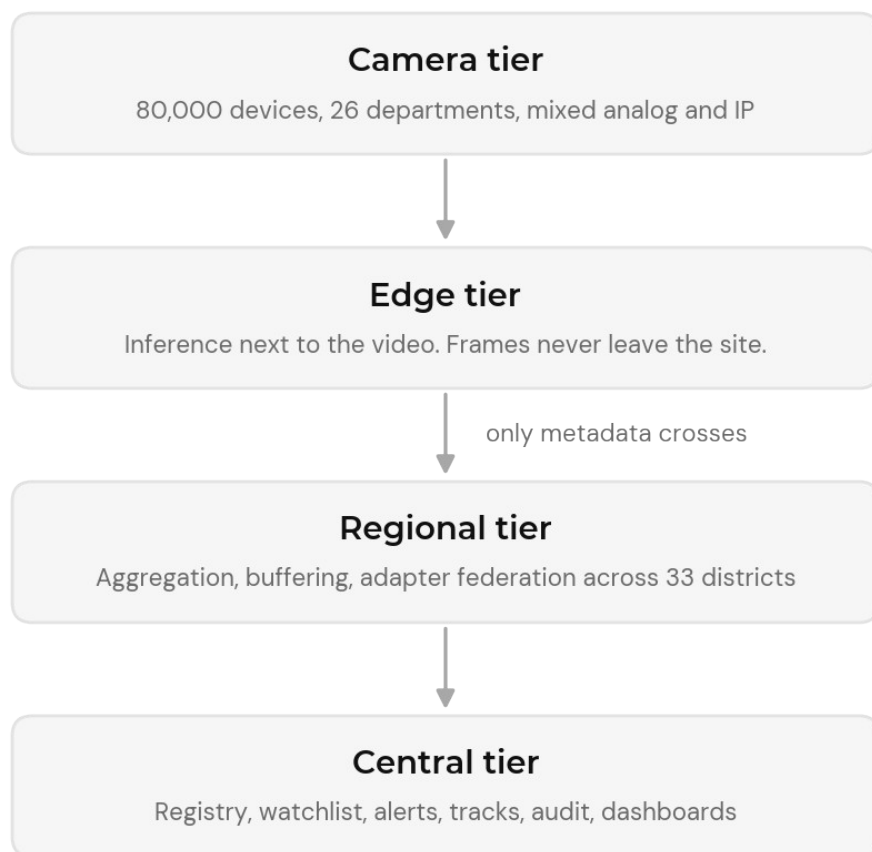


Figure 1. The four tiers. The labelled edge is the single most important line in this design.

The line between the edge tier and the central tier is the one that matters. Frames never cross it. What crosses is a class name, a confidence, a bounding box, a timestamp, provenance, and where ANPR is enabled and permitted, plate characters. The central API therefore carries no computer-vision dependency at all, which keeps it small enough to deploy anywhere and means a compromise of it yields no imagery.

2.2 Components and stack

Component	Stack	Status
central-api	FastAPI, SQLAlchemy 2 (async), PostgreSQL	Running
Adapters	One per source system, common abstract base class	Running, 3 sources
edge-worker	Python, Ultralytics YOLO, PaddleOCR, OpenCV	Running
Consensus ANPR	Vendored anpr/ package, per-track voting	Running
Incident detection	Derived from tracker boxes, no additional model	Running
Dashboard	Next.js 14, React 18, Tailwind, Leaflet with OpenStreetMap	Running
Storage	PostgreSQL with JSONB; SQLite for the test suite	Running

Table 3. Implementation status of every component in the platform.

Every dependency is open source. There is no vendor account, no API key and no licensed component anywhere in the stack. The map is OpenStreetMap and the models are open weights. Everything is containerised, and the whole demonstration stack starts with a single compose command.

The edge worker has three build targets so a site pulls only what it will run: a base image of roughly 120 MB carrying a mock detector for continuous integration, a YOLO image of roughly 2.5 GB for real inference, and an ANPR image of roughly 3.5 GB carrying the consensus reader. Model weights are mounted, never baked in.

2.3 Component interactions

Three interaction paths carry essentially all of the platform's behaviour. Each is listed with the components it touches, in order.

Feature	Details
Onboarding and federation	• An installer completes the CCTV installation form in the owning department's own system, where it is held as a draft. • That department validates it: required fields, coordinates inside its jurisdiction, no duplicate device ID, a non-empty local-role list. There is no approval stage. • On success the camera is registered. Vigentra's sync service polls each adapter, which normalises the department dialect into the canonical camera model, and writes it to the registry. • Only registered and synchronised records enter the central registry. Suspended and decommissioned assets are retained and marked unavailable, so a withdrawn camera never silently disappears.
Detection, correlation and alerting	• The edge worker captures frames, runs object detection, and where enabled runs the consensus ANPR engine over each tracked vehicle. • It posts detections and plate reads to the ingest endpoint as metadata. No frame is transmitted. • Ingest creates a sighting for each plate read, and the matcher compares it against the active watchlist inside the same transaction. • A match raises an alert, which reaches the console and the audit log. The track service assembles sightings into a route on request.
Authorised viewing	• An operator requests a session for a camera. The permission service evaluates account, role, camera state, ownership or grant, and city and zone scope, in that order. • A permitted session returns an opaque Vigentra address. The browser never learns the upstream host. • Every media segment re-enters the same route and is re-authorised, so a revoked grant stops a playing stream within one segment. • Every outcome is audited, including the refusals.

3. Integrating heterogeneous cameras, NVRs and VMS

3.1 The adapter is the whole answer

One adapter class per department system is the only component in Vigentra that knows a vendor’s field names, authentication scheme, lifecycle vocabulary, timestamp format or error envelope. Above the adapter boundary, everything is canonical. Two mock department systems exist specifically to prove this against genuinely incompatible dialects rather than against a convenient one, and a third source is the live government sandbox grid.

	Traffic Police	Municipal Corporation	Sentinel grid
Field style	flat snake_case	nested camelCase	flat, minimal
Timestamps	IST wall-clock strings	epoch milliseconds	absent
Authentication	X-API-Key header	Bearer token	credentialed URL
Installation register	Yes	Yes	None
Recorded archive	Yes	Yes	None

Table 4. Three source systems with deliberately incompatible surfaces, all normalising to one canonical camera model.

The adapter contract is a small, fixed set of methods covering the installation lifecycle, the approved-camera listing, per-camera health and a source health check. Onboarding a new vendor is three steps: implement the contract, add a configuration entry,

synchronise. Nothing downstream changes, and the dashboard cannot tell the sources apart.

3.2 Where the NVR sits, and why Vigentra never touches it

The adapter contract deliberately defines no method that hands Vigentra a stream URL, an NVR address, a credential or a long-lived media token. The owning department holds its NVR recordings, its RTSP addresses, its camera credentials and any local viewing tokens, and it decides whether Vigentra may broker each camera at all. That switch is not overridable by any Vigentra role.

Recorded playback therefore works by request rather than by access. An operator asks for a date and a time window; the request goes to the owning department, which returns the segment of its own recording covering that window as a media fragment. Vigentra never mounts, indexes or copies the NVR. What bounds playback is the owning unit and the camera's own retention period, not a central quota, and footage older than that retention is gone rather than withheld, which the refusal states explicitly.

3.3 Integration routes by what a department exposes

What the department exposes	Integration route	Effort
A documented VMS API or SDK	A new adapter implementing the standard contract, plus a configuration entry.	Preferred
ONVIF Profile S or G endpoints	Adapter over ONVIF device and media services for discovery and stream profiles, with recording retrieval over Profile G where present.	Supported
RTSP paths only, no management API	Registry populated by bulk CSV onboarding, feed access by RTSP path convention. Health degrades to reachability rather than vendor status.	Workable

What the department exposes	Integration route	Effort
Analog cameras behind an encoder or DVR	The encoder or DVR is the integration point and presents as an IP source. The analog leg is invisible above it.	Workable
No network exposure at all	Registry-only federation. The camera is listed, mapped and counted in gap analysis, with no feed.	Registry only

Table 5. Integration routes. Every department can reach at least registry-level federation on day one.

Three adapters are implemented today, all on the first route: two mock department systems and the live government sandbox grid. The remaining rows are designed routes rather than built ones, and each is named that way deliberately. What makes them cheap is that none of them changes anything above the adapter boundary.

REGISTRY FEDERATION DOES NOT WAIT FOR FEED INTEGRATION

A department with no API, no ONVIF and no reachable ports can still be federated from a spreadsheet on the day it provides one. Its cameras appear on the map, in coverage-gap analysis and in the statewide inventory immediately. Feed integration is a separate, later step, and nothing else is blocked behind it.

3.4 Three properties that took deliberate work

Feature	Details
Read-only sources are read-only in code	- The grid has no installation register, so every write in its adapter refuses locally rather than issuing a request that should never be made. - Listing installation requests returns an empty list because the register genuinely does not exist, not because a call failed.

Feature	Details
Capability is enforced twice	• Grid cameras advertise live and never playback, and the video adapter refuses a playback request outright. • Two independent gates, because the absence of a recording should not depend on either one alone.
A source outage is isolated	• One department going down leaves the others fully working. Its cameras go offline with a stated reason. • Listings degrade to the last mirrored state and say so through a response header, and every failed poll is written to the audit log. • Demonstrable by stopping one department service and refreshing the overview page.

4. Ingesting and managing live streams from dispersed sites

Camera sites run from dense urban Ahmedabad to border districts a thousand kilometres out, and the connectivity is not uniform. The design assumption is that a site will lose its uplink, not that it might.

4.1 Capture for inference

The edge worker captures live directly, following the sandbox integrator's guide. Every rule in that guide is enforced in code and pinned by a test that runs against a fake capture, so the rules hold whether or not the sandbox is reachable. That matters more than it sounds: a rule only checked when the feed is up is not really checked.

Rule	Implementation
Force RTSP over TCP	Set as a process-wide transport option before the first capture is constructed.
Never trust the reported frame rate	No code path reads the declared FPS property. Measured rate is derived from presentation timestamps; the catalogue value is stored as provenance only.

Rule	Implementation
Drive timing from PTS	Frame timing comes from the presentation timestamp, never from arrival time.
Tolerate inter-frame gaps	A gap is the real elapsed PTS delta, and does not tear down the capture.
Reconnect with backoff	Exponential backoff from 2 seconds to a 30 second cap, never a tight loop.
Tolerate join-time decoder noise	Twenty-five bad reads tolerated after each connect or reconnect.
Expect scene discontinuity	A backwards PTS jump beyond 5 seconds flags a discontinuity and resets long-lived state; a smaller backwards step is treated as timestamp jitter.
Write nothing to disk	Frames are decoded from a live capture and never persisted.
Never publish to the gateway	No write verb appears anywhere in the capture module.
Pace the load	Catalogue cached on a short TTL, one capture per camera, released on exit.

Table 6. Integrator guide rules, each enforced in code and asserted by a test.

THE RULE THAT SILENTLY RUINS A TRACKER

The gateway replays a buffered group of pictures on connect, so the first second arrives faster than real time. Anything timestamping by arrival computes impossible velocities after every reconnect. Driving timing from the presentation timestamp is what prevents it.

4.2 Operating over unreliable and constrained links

Feature	Details
Reconnection, already built	- Exponential backoff from 2 seconds to a 30 second cap, tolerating join-time decoder noise and surviving feed loops. - Every rule is pinned by a test running against a fake capture.
Transport fallback, already built	- RTSP on port 8554 is filtered on many networks. The worker probes and falls back to HLS on 443, which is the path that works everywhere. - Single-camera viewing is HLS for the same reason.
Sampling as a bandwidth dial, already built	The frame sample interval trades detection density for load, tunable per site without a redeploy.
Edge buffering, designed and half built	- Detection IDs are deterministic from camera, instant, class and box, and ingest is idempotent on them, so a replay collapses rather than double-counting. That is the hard part, and it is built and tested. - The local queue that holds detections while the uplink is down is not yet built. Named here rather than implied.

The regional tier exists for exactly this reason. Aggregating roughly 33 districts means a site talks to something nearby rather than to a single central endpoint a thousand kilometres away, and a regional outage degrades one district rather than the state.

4.3 Brokered video for human viewing

Inference pulls frames at the edge. Human viewing is a separate path, and it is brokered so the browser never learns the upstream host exists.

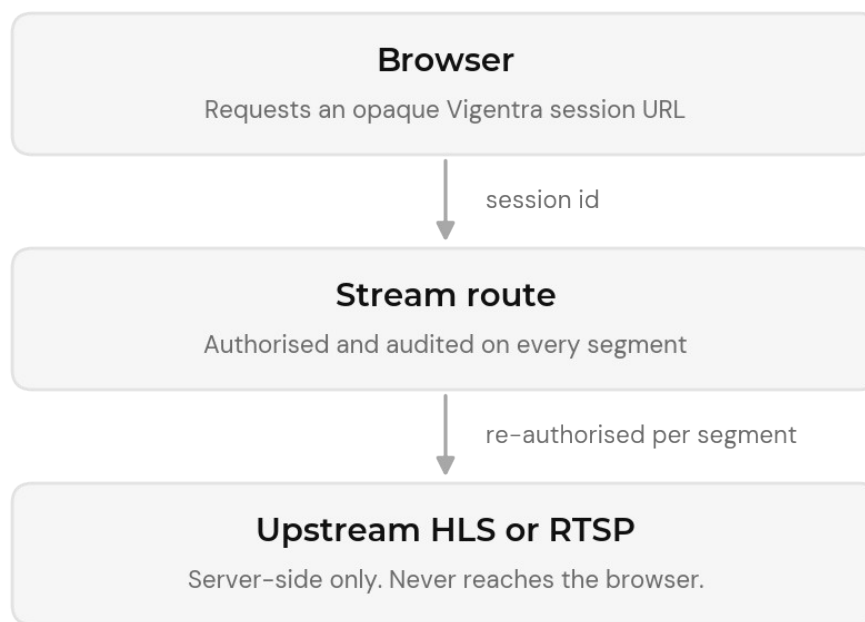


Figure 2. Brokered video. Authorisation is re-checked on every segment, not only at session open.

Every URI in every playlist is rewritten, both bare lines and URI attributes on map, part and key tags, to a relative reference. Relative is deliberate: the rewrite stays correct behind any reverse proxy without the service needing to know its own public prefix. Because every segment re-enters the same route, authorisation is re-checked per segment, so revoking a grant stops a playing stream within one segment rather than at session expiry.

The path reference arrives from the browser and is treated as hostile. A scheme, an authority, a leading slash or a parent-directory segment is refused rather than normalised, because a permissive resolver there would turn the broker into an open forward proxy.

VIEWING MANY CAMERAS AT ONCE IS A LOAD PROBLEM, AND IT WAS MEASURED

The sandbox HLS gateway delivers a 6-second segment in 15 to 80 seconds and refuses overlapping requests, so a browser wall of fifty tiles blacks out entirely. The live wall is therefore served by an edge frame service. It decodes each feed in a bounded pool, draws the edge's vehicle detection on every frame and, on chosen feeds, runs the full ANPR engine live. A tile is answered from memory in about two milliseconds, and holds a session only while it is on screen.

5. Watchlist integration and continuous correlation

5.1 What a watchlist entry is

The platform implements the five watchlist categories the challenge names, as first-class enumerated values rather than free text.

Category	Intended use
stolen	Vehicles reported stolen.
wanted	Vehicles associated with a wanted person or an active case.
blacklist	Vehicles barred on standing departmental instruction.
missing	Vehicles associated with a missing-person case.
suspect	Vehicles under suspicion in an open investigation.

Table 7. Watchlist categories, implemented as an enumerated type.

Each entry carries the registration, its category, a mandatory non-blank reason, an optional case reference, the username that added it, the owning department, an active flag, an optional expiry, and a running count of the alerts it has produced. That last field is operational rather than decorative: a plate firing constantly usually sits one confusion pair away from something common, and the count is what surfaces it.

Feature	Details
An entry needs a reason	Mandatory, non-blank, recorded with the username. A standing instruction to flag a vehicle statewide that nobody can account for is the one that should never have been added.

Feature	Details
An entry should have an end date	• Optional but pressed for in the interface, and expiry is applied on read rather than by a purge job, so a job that fails to run cannot quietly keep a vehicle flagged for ever.
Entries are stood down, never deleted	• Deactivation records who and why. The trail is the point.

5.2 Correlation happens at ingest, continuously

Every plate the edge reads becomes a sighting: one vehicle, read once, at one camera, at one instant. It is stored separately from the detection that produced it, because a detection is an observation of an object and a sighting is an assertion about an identity. Different retention, different permissions, different consequences when wrong.

Each sighting is matched against the active watchlist inside the ingest transaction, not at query time. An alert asserts that the system knew at a moment in time, and a query-time matcher cannot make that claim. It also means a watchlist entry added after a vehicle passed does not manufacture an alert for that pass, which is correct and is the surprising half of the behaviour, so it is tested explicitly.

5.3 Why matching is fuzzy, and how it is priced

Exact-match lookup over OCR output is a trap. If the only camera that saw a vehicle read one character wrong, an exact query returns nothing and the vehicle looks like it was never there. Plain edit distance is too blunt, so operations are priced by what the reader actually gets wrong.

Operation	Cost	Reasoning
Known confusion pair, B against 8, O against 0	0.35	What the reader actually gets wrong.
Unrelated substitution	1.0	Usually a different vehicle.
Insertion or deletion	0.5	A character lost to glare or a frame edge is commonplace. A different digit in its place is not.

Table 8. Confusion-weighted edit distance. The default threshold is 1.0, about two plausible OCR errors.

Raising the threshold does not find more stolen vehicles. It finds more innocent ones. The distance that admitted each hit is recorded on the alert, and exact is carried as its own flag, because an operator acts differently on the two.

5.4 Integrating external watchlist databases

The watchlist today is populated through the platform's own authenticated endpoints, with every addition and stand-down audited. Integration with a departmental or national source, VAHAN, SARTHI, eGujCop, AFIS or NAFIS, is designed for and is not connected.

THE OFFICIAL-SOURCE PROVIDER REFUSES RATHER THAN GUESSES

A provider interface exists as a documented shape and is inert until an authorised URL and token are configured. It does not fall back to synthetic data, and it does not approximate. Every demonstration record in this platform is synthetic and labelled as such. When an authorised source is configured, it enters through the same permission, retention and audit machinery the watchlist already uses, rather than beside it.

6. AI-powered video analytics

Five analytics are addressed. Three are built and running, one is built and deliberately conservative, and one is not implemented, with the reasoning given in full rather than left as an omission.

Analytic	State	Where it runs
Object detection	Running	Edge, continuous, all cameras
ANPR, consensus engine	Running	Edge, off by default, permission-gated
Vehicle and person tracking	Running, per camera	Edge tracker; cross-camera by plate only
Incident detection	Running, candidates only	Edge, derived from tracker boxes
Facial recognition	Not implemented	Section 6.5 states the position

Table 9. Analytics status. Section 6.5 addresses facial recognition directly.

6.1 Object detection

YOLO11n runs at the edge over a canonical vocabulary: person, car, motorcycle, bus, truck, auto-rickshaw and bicycle. COCO classes outside that mapping are discarded rather than guessed at. A frame-quality router classifies each frame first and skips ones too degraded to mean anything, recording that it did so rather than producing noise.

6.2 ANPR: a consensus engine, not a per-frame OCR call

Number-plate reading is the most sensitive capability the platform has, and it is also the one that determines whether the graded test case works at all. It is worth being precise about what it does.

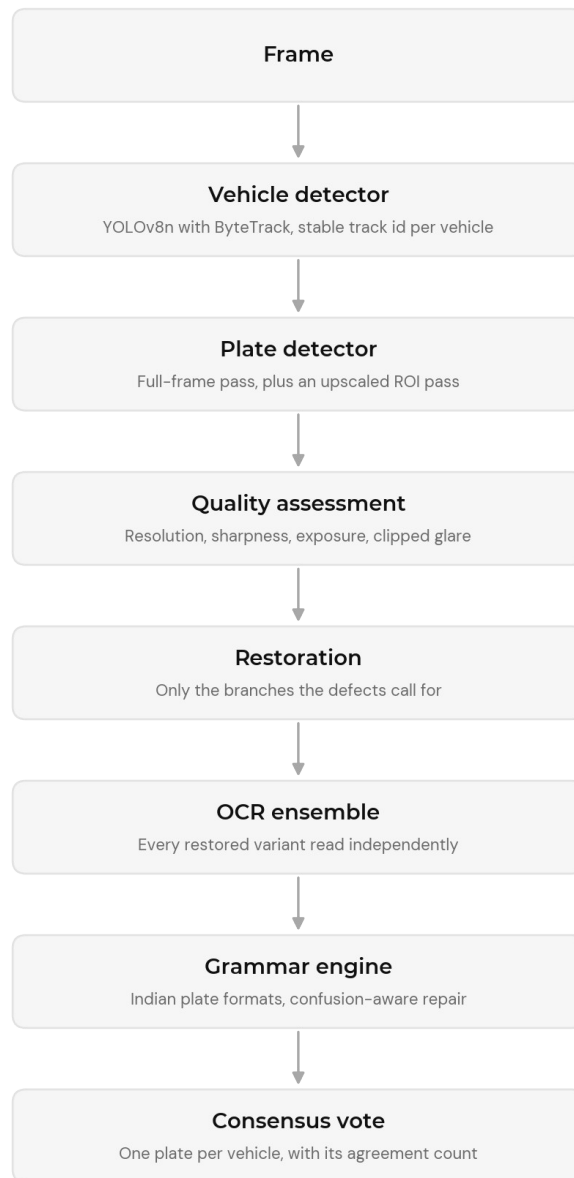


Figure 3. The consensus ANPR pipeline. Restoration branches include perspective rectification, CLAHE and unsharp, a low-light branch, a glare branch, four-times super-resolution for small plates, and adaptive binarisation.

The two layers carrying the accuracy are the grammar engine and the consensus vote, not the OCR model.

Feature	Details
Grammar	- Indian plates have rigid formats, so a character's position says whether it must be a letter or a digit. - GJ05JV34S6 is unambiguously GJ05JV3456, because slot eight cannot hold a letter. - Repair is a bounded search over one-to-many confusion sets, constrained by the format regex and the real state-code list, picking the cheapest fix. That is how OL8CAF5030 recovers to DL8CAF5030. - Formats covered: standard, Delhi single-digit district, Bharat series, diplomatic, and older short-tail plates. A soft Gujarat prior breaks ties without preventing other states validating.
Consensus	- A plate is emitted once per vehicle, with the count of frames that agreed. - That count travels to the operator's screen, because one frame is a guess and twelve frames agreeing is a reading, and an operator is entitled to know which they are looking at.

Anything that does not parse as a plausible Indian registration is dropped at the edge and never transmitted, then checked again centrally. Half-read text is worse than no text, because it looks like evidence and is not. An operator acting on a misread digit is the failure mode worth engineering against: a missing plate costs a lookup, a wrong one costs someone a knock on the door.

6.2.1 Why a camera reads nothing

Silence used to be ambiguous. A camera reporting no plates might have seen no traffic, or might have seen two hundred vehicles whose plates were forty pixels wide. Those are opposite situations with opposite fixes. Every vehicle that leaves the frame now settles into one of three outcomes.

Verdict	Meaning	Whose problem
CONFIRMED	The vote settled and the restorations agreed.	None
UNCERTAIN	The plate was big enough to read; the readings disagreed.	This vehicle
UNREADABLE	The plate never reached a size any recogniser resolves.	This camera's placement

Table 10. Per-vehicle readability verdicts. Only UNREADABLE is a statement about the camera.

Size bands are shared between the running pipeline and the camera suitability survey, so the two cannot drift apart. A plate is read off the unmodified crop at 160 pixels wide or more, recovered by restoration at 120, read only when several restorations agree at 80, attempted with agreement deciding at 40, and not attempted below 40.

THE 40 PIXEL GATE MUST NOT BE RAISED, AND MUST NOT BE LOWERED

Plates on this estate have been read correctly by eye at 53 to 90 pixels, and no width separates the legible ones from the illegible: a 71 pixel motion-blurred plate is unreadable while a 53 pixel sharp one is not. Raising the floor discards real evidence. Lowering it is worse: undersized crops do not come back as near misses, they come back as fabrications. GJO6D02415 read as LD607415, and GJO1MR4873 as GI667673. A fabricated registration is a wrong vehicle attached to a real place and time.

6.3 Person and vehicle tracking

Tracking happens at two levels, and the distinction between them is a deliberate limit rather than an incomplete feature.

Scope	Vehicles	People
Within one camera	ByteTrack, stable track id, used for consensus voting and incident detection	Tracked within the frame, used for incident detection only
Across cameras	By plate read only	Not performed
By appearance	Not performed	Not performed

Table 11. Tracking scope. Cross-camera association is by registration number and nothing else.

Given a registration number, the platform returns every sighting the caller is permitted to read, ordered by time, with per-leg distance, elapsed time and implied speed, drawn on a map. Four behaviours are worth stating, because each is a decision.

Feature	Details
Built from plate reads only	- No appearance matching, no re-identification by colour or shape, no join to the vehicle reference registry. - A vehicle whose plate was not read contributes nothing to its own route. That understates movement rather than inventing it, and that is the right direction to be wrong in.
Repeated reads at one camera collapse into one pass	A vehicle waiting at a signal would otherwise produce a dozen points and a route that looks like frantic activity in one spot.
Impossible legs are flagged, never dropped	- A leg implying more than 200 km/h usually means one of the two reads belongs to a different vehicle. - That is a finding about the route's reliability, surfaced in the API, the table and the map. Removing it would make the line look cleaner and be less true.
Scope is applied before assembly	- A route is never built from a sighting the caller could not have read. - Filtering afterwards would leave holes that look like the vehicle disappeared rather than like permissions ending.

Every trace requires a written reason, enforced at the API rather than in the interface, and recorded against the account with the plate and the time.

6.4 Incident detection

Five traffic events are watched, all derived from the tracker's own boxes over time. No additional model runs, so this costs almost nothing on top of detection that is already running.

Event	Severity	What triggers it
PERSON_ON_CARRIAGEWAY	HIGH	A person in the path of moving traffic.
COLLISION_CANDIDATE	HIGH	Two vehicles overlapping in the image plane while both lose speed sharply together.
SUDDEN_STOP	MEDIUM	A vehicle decelerating far harder than the traffic around it.
WRONG_WAY	MEDIUM	A vehicle travelling against the camera's established flow.
STOPPED_IN_LANE	LOW	A vehicle stationary while other vehicles keep moving past it.

Table 12. Incident types and their severities, used directly for alert prioritisation in section 7.

Feature	Details
Speed is measured in vehicle heights per second, not pixels	- A pixel does not mean the same thing across the frame. On one grid camera a vehicle at the top of the image is 40 pixels tall and the same vehicle near the lens is 300, so a fixed pixel threshold fires constantly in the near field and never in the far field. - Dividing by box height removes most of the perspective term. It is not a calibration and yields no real-world speed. It makes one threshold behave roughly the same across the frame, which is all that is claimed.

Feature	Details
Direction of flow is learned per camera, not configured	- Wrong-way detection needs to know which way traffic goes, and that differs per camera and cannot be hand-set for dozens of them. - The detector accumulates the median heading of confirmed tracks and only begins judging direction after it has seen enough of them. Until then it reports nothing rather than guessing.
One event per track, not one per frame	A per-track-and-kind cooldown prevents a single event being reported repeatedly while it is still happening.

INCIDENT DETECTION PRODUCES CANDIDATES, NEVER FINDINGS

This module shares ANPR's governing rule. ANPR must never invent a registration; incident detection must never assert an accident. There is no accident footage from this estate, so none of the thresholds has been validated against a real collision, and they are reasoned rather than measured. A module announcing a confirmed accident on that basis would be claiming a confidence nobody has earned. Every output is a candidate with its evidence attached, and every threshold should be re-derived the first time a real incident is recorded here.

6.5 Facial recognition: the position, stated directly

Facial recognition is named in the challenge brief and is not implemented in Vigentra. Leaving that as a silent omission would be the wrong answer to a graded requirement, so the reasoning is given in full.

Feature	Details
The optics on this estate do not support it	- Section 10.3 measures a median plate width of 32 pixels across 37 feed runs, with zero consensus-confirmed reads. - A face carries far less contrast and far more variation than a retro-reflective plate with a fixed character set and a rigid grammar. An estate that cannot resolve a plate cannot resolve a face, and no model choice changes that. - Claiming facial recognition on this footage would be claiming the one thing section 10 spends its length refusing to claim.

Feature	Details
It is a different category of decision, not a larger version of ANPR	- A registration number identifies a vehicle and, through a registry, a person. A face identifies a person directly, and is not revocable, not reissued, and not something anyone can decline to carry. - The platform already separates the operational fact from the identifying one, and gates plate reading behind its own permission, its own shorter retention clock and audited disclosure. Facial recognition would need at least that, plus a stated legal basis and a governed gallery, neither of which is a technical artefact this team can supply on its own.
The architecture has the slot, if a department requires it	- Analytics run at the edge and only metadata crosses to the centre. A face matcher would be another edge analytic emitting a match reference rather than an image, exactly as ANPR emits characters rather than a crop. - It would carry its own permission, separate from plate reading and held by no role by default. It would carry its own retention clock, shorter than plates rather than longer. Every disclosure would be audited the way a plate disclosure already is. - What it would additionally require: cameras sited and specified for face capture, a lawfully held and governed gallery, and a written authority for its use. Those are procurement and policy inputs, and the platform should not pretend to supply them.

WHAT THIS MEANS FOR THE WANTED-PERSON AND MISSING-PERSON CATEGORIES

Both categories are implemented and both match on registration number today, against a vehicle associated with the case. They do not match on a face. An alert in either category asserts that a watchlisted vehicle was seen, never that a person was identified, and the console says so on the surface that shows it.

7. Alert generation and notification workflow

7.1 From a read to an operator

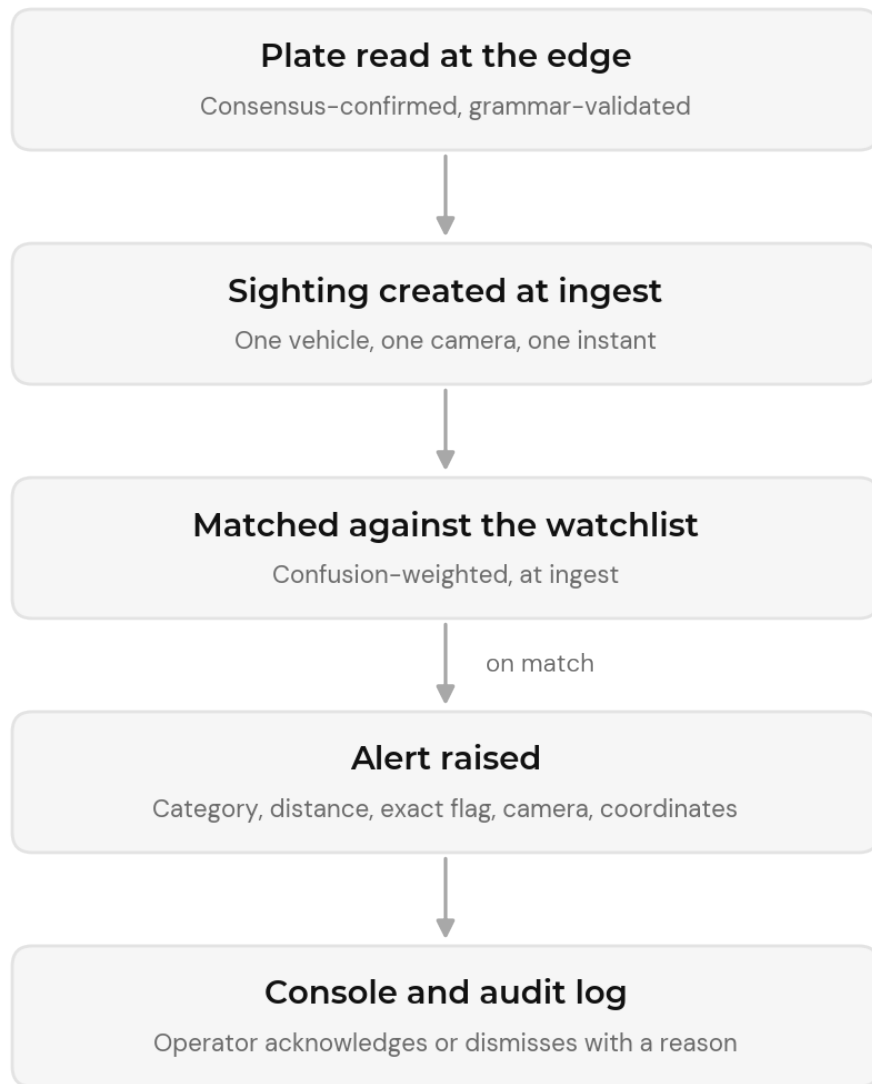


Figure 4. Alert lifecycle. The matcher writes its own audit entry, because the trail has to show what the system concluded as well as what people did.

7.2 Prioritisation

Alerts are not a flat queue. Four signals order them, and each is carried as its own field so an operator can sort and filter rather than accept one opaque score.

Signal	Values	How it is used
Match quality	exact, or a distance above 0	Exact matches act on directly. Near matches are flagged for review, with the distance shown.
Category	stolen, wanted, blacklist, missing, suspect	Carried on every alert so an operator can filter to the categories their unit acts on.
Acknowledgement state	open, acknowledged, dismissed	The console opens on unacknowledged alerts by default.
Incident severity	HIGH, MEDIUM, LOW	Applies to incident candidates rather than watchlist alerts. Person on carriageway and collision candidate are HIGH.

Table 13. Prioritisation signals, each a separate field rather than a single blended score.

NEAR MATCHES ARE SHOWN BY DEFAULT, AND THAT IS THE IMPORTANT DEFAULT
 The filter to hide them exists and ships switched off. Suppressing near matches keeps a console tidy, and it is also exactly how a vehicle whose plate was read one character wrong passes a camera and nobody hears. The console instead shows a running count of alerts needing review because they are near rather than exact.

7.3 Visualisation

Feature	Details
The alert console	• A table of open alerts, each row carrying the category, an exact or near marker, the distance, the camera, the district and city, and the time. • A header count of alerts needing review because they are near matches rather than exact. • Filters for unacknowledged only, on by default, and hide near matches, off by default, plus a time window.
The map	• Camera markers with status colouring, alert pins at the camera that raised them, and a route polyline connecting a traced vehicle's sightings in time order. • Legs implying an impossible speed are drawn in red rather than removed. • OpenStreetMap tiles through Leaflet. No API key, no vendor account.
What an alert never looks like	• An alert is never presented as an identification. It is a probabilistic reading of a photograph matched fuzzily against a list, and the console says so on the surface that shows it. • An account without plate permission still sees that an alert of a given category fired at a given camera, with the registration withheld rather than the row refused.

7.4 User interaction and workflow states

Watchlist alerts and incident candidates have different workflows, because an alert asks whether to act and a candidate asks whether it is real.

Object	States	Actions and record kept
Watchlist alert	open, acknowledged, dismissed	An operator acknowledges, or dismisses with a reason. The username, timestamp and reason are stored on the alert and written to the audit log.
Incident candidate	CANDIDATE, REVIEWING, CONFIRMED, DISMISSED	A reviewer moves it through the states, and the reviewer, review time and note are stored against the candidate with its evidence.

Table 14. Workflow states. Neither path allows an automatic action.

Acknowledgement and dismissal are separated deliberately. Acknowledging records that a human saw it; dismissing records that a human saw it and judged it not actionable, and asks why. A console where the only action is to make a row disappear produces no evidence about whether the system is useful.

7.5 Notification today, and what is not built

The console re-checks for new alerts every 30 seconds and shows when it last refreshed, and the camera health monitor polls every department on its own configured interval. That is the notification mechanism today: an operator watching a console sees a new alert within one refresh.

THERE IS NO PUSH NOTIFICATION, AND IT IS NOT CLAIMED

No webhook, no server-sent events, no SMS or email dispatch, and no mobile push exists in the platform. For a control room with a console permanently open, polling is adequate and honest. For an officer in the field it is not, and that gap is named here rather than described as though it were implemented. The seam for it already exists: the matcher raises alerts through a single call over a list of plate reads, which is the same seam section 8.4 identifies for moving alert fan-out onto a message topic at scale.

8. Scalability, interoperability, security and performance

Two things stated first, so the rest is read correctly. Vigentra federates 30 live sandbox cameras today plus two mock department systems, and everything at that scale is measured. Everything above it is engineering estimate and is labelled as such. The edge tier scales horizontally today with no design change, which is a property of putting inference next to the video and shipping only metadata. The central tier has three specific things that do not scale as written, and they are named in section 8.5 rather than hidden.

8.1 Why bandwidth is not the binding constraint

	Central video streaming	Vigentra metadata crossing
Per camera	about 2 Mbps, H.264 1080p at 15 fps	about 48 kbps at peak, 400 bytes per detection
80,000 cameras, peak	160 Gbps sustained	about 3.8 Gbps
Realistic mixed load	160 Gbps, it does not vary	600 Mbps to 1 Gbps
Per day	about 1.7 petabytes	metadata only, no video stored

Table 15. Bandwidth arithmetic. The difference is a factor of 40 to 250.

160 Gbps sustained is not a network anyone is going to build, and it is why a fully central VMS is the most expensive of the reference models to operate. The trade Vigentra makes is that a central operator cannot arbitrarily scrub any camera's history. That is the correct trade, because that capability is also the one with the worst privacy properties. Video is still available: it is pulled on demand, per session, for the cameras someone has a reason to watch.

8.2 Edge compute

Measured: YOLO11n object detection runs at roughly 190 milliseconds per frame on CPU. Sampling every fifth frame of a 25 fps feed is 5 inferences per second per camera.

Estimated: a single mid-range T4-class accelerator running batched YOLO11n at 640 pixels handles roughly 150 to 200 inferences per second, so about 30 cameras per GPU for object detection. ANPR is heavier, the ROI pass plus several restoration variants plus OCR, and realistically halves that to about 15 cameras per GPU.

Scenario	Accelerators, estimated
Object detection everywhere	about 2,700
ANPR everywhere as well	about 5,300
ANPR only where the optics allow it	about 3,000

Table 16. Edge accelerator estimates for 80,000 cameras. These are planning estimates, not measurements.

The third row is the recommendation, and the estate survey in section 10.3 is what justifies it. It is not a cost dodge, it is what the optics allow. Running ANPR on a general-purpose overview camera spends an accelerator to produce almost nothing, which the survey measured directly: zero confirmed reads across an estate whose median plate is 32 pixels wide.

So cameras are classified on onboarding. The Model 1 registry already holds installation purpose, camera type and view direction for every camera, and the grid reference data already distinguishes red-light violation units from general PTZ and bullet cameras. That classification drives which analytics run where, as an operational decision made once at onboarding rather than a blanket setting.

At Indian public-procurement pricing, 3,000 T4-class accelerators plus host hardware is on the order of 90 to 120 crore rupees for the edge tier, deployed over the rollout period rather than at once. That is an estimate for planning and not a quotation.

8.3 Storage

Vigentra stores no video. Departments keep their own footage under their own retention policies, which already differ and which the federation does not need them to reconcile. What Vigentra stores is metadata, tiered.

Tier	Contents	Retention	Store
Hot	Detections, sightings, alerts, health	30 days	PostgreSQL with TimescaleDB, partitioned by day
Warm	Hourly aggregate counts per camera; sightings only for plates that hit the watchlist	1 year	Same cluster, compressed hypertables
Cold	Audit log, watchlist history	7 years	Object storage, append-only

Table 17. Proposed metadata retention tiers.

Plates keep their own shorter clock, 30 days by default, already enforced on read as well as by any purge job. The audit log is the one thing that never ages out on a shortened clock. It is append-only, it is what makes every other control checkable, and it is small: one row per act, not per frame.

8.4 Central tier

Concern	Approach
Ingest throughput	About 1.2 million detections per minute statewide at peak. Batched inserts, partitioned tables, and a write path that touches no shared row.
Read and write split	Dashboards, reports and route queries go to read replicas. Ingest goes to the primary.

Concern	Approach
Alert fan-out	The matcher runs in the ingest transaction today. Past a few thousand cameras it moves to a message topic between ingest and matching. The interface is already a single function call over a list of plate reads, so this is a substitution rather than a rewrite.
High availability	Primary with synchronous standby per region, and point-in-time recovery to object storage.
Disaster recovery	RPO 5 minutes, RTO 1 hour. The registry is reconstructible from the department systems by re-running sync, which is a real advantage of the federation design: the authoritative copy of every camera record still lives with its owner.
Orchestration	Kubernetes, one namespace per tier, autoscaling on ingest queue depth rather than CPU.

Table 18. Central tier scaling approach.

8.5 What does not scale as written

Three things. Naming them is more useful than a diagram that implies they are solved.

Feature	Details
Fuzzy plate search is linear in Python	- Candidates are narrowed by time and scope in SQL, then confusion-weighted edit distance runs row by row. Comfortable at sandbox scale, not at statewide scale. - Replacement: a trigram index as a cheap prefilter, cutting candidates by two or three orders of magnitude before the weighted distance runs on what survives. The weighted distance gives the ranking its quality and should be fed by the prefilter, not replaced by it.

Feature	Details
The watchlist matcher loads the full active list per batch	• Cached briefly and invalidated on write, which is right for a list of hundreds. For hundreds of thousands it becomes a per-batch scan. • Replacement: a bloom filter over exact plates for the common negative case, plus the trigram prefilter for near matches. Exact hits are the majority of real alerts and should not pay for fuzzy matching at all.
Coordinates are plain floats, not PostGIS	• Fine for the map rendered today. Nearest-camera, polygon-containment and route-corridor queries all get materially harder without it, and those are exactly the queries a mature tracking feature wants. • Replacement: PostGIS with a spatial index on a geography point column. This is the organisers' own suggested stack, and it is a migration rather than a redesign because every write already goes through one mapper.

8.6 Interoperability

Layer	How openness is preserved
Source systems	One abstract adapter contract with a canonical error taxonomy. No vendor-specific logic exists anywhere above it, including in the console.
Camera and stream	RTSP and HLS, with ONVIF as the discovery and profile route where a department exposes it. No proprietary SDK is required by the platform itself.
Platform API	REST over JSON with an OpenAPI description, so a department or a future system integrates without a client library from us.
Mapping	OpenStreetMap through Leaflet. No API key, no vendor account, no usage tier.
Models	Open weights, mounted rather than baked into images, so a department can substitute a model it has validated itself.

Layer	How openness is preserved
Data	A documented canonical camera schema, and bulk CSV onboarding that takes a spreadsheet as its input format.

Table 19. Interoperability by layer. There is no licensed component anywhere in the stack.

8.7 Security

Metadata and footage were once gated together, and answering both questions with one gate was the mistake. A Traffic officer could not learn that a Municipal camera covered the junction they were investigating, which made the central registry useless for the case it exists to serve while doing nothing for privacy. Knowing a camera is mounted at a public junction is not sensitive. The footage is, and the footage was behind the same single gate. They are now separate.

Control	Implementation
Credentials never centralise	Vigentra stores no RTSP URL, no NVR address, no credential and no media token. A breach of the central tier yields no key to any camera.
Sessions are brokered and opaque	Short-lived, watermarked with username and timestamp, bound to the operator who opened them, re-authorised per segment.
Three-dimensional scope	Department, city and zone, checked per request, plus the camera's own owner policy.
Cross-unit access is asked for	Personal, time-boxed and revocable grants with a mandatory reason. The operator who runs a unit's cameras is the approver, so there is no separate approval role to route around.
Least privilege, finely split	13 roles across 22 permissions. Plate reading, watchlist reading, watchlist management, alert reading, alert acknowledgement and track reading are all separate permissions.

Control	Implementation
Field-level redaction	Three visibility depths. Cross-department reads are capped so a unit's internal operational data stays its own, and withheld fields are returned by name rather than as blanks.
Withheld, not refused	An account without plate permission still gets the alert, flagged as plate withheld. The operational fact and the identifying fact are separable, so they are separated.
Retention enforced on read	Plates carry a shorter clock, 30 days by default, applied at read time as well as by any purge, so a failed purge cannot quietly extend availability.
Complete audit trail	Every onboarding step, sync, metadata read, footage request, decision, session, refusal, plate disclosure, watchlist change, alert and trace. Rows commit independently of the request that produced them, because a denied attempt is the record most worth surviving.
Machine decisions audited too	The alert-raised entry is written by the matcher, not by a person. The trail shows what the system concluded as well as what people did.

Table 20. Security controls, each implemented and covered by the automated test suite.

THE PERMISSION SPLIT THAT MATTERS MOST

The edge account that ingests plates holds none of the read permissions. A worker that can read the watchlist back is a worker that can exfiltrate it. Municipal roles hold no watchlist, alert or track permission at all, because civic monitoring counts vehicles and does not identify their owners.

A refusal carries a state that distinguishes the kinds of no: another unit owns it and should be asked, the owner has brokered video off for this camera and there is nothing to request, the camera is unavailable, or role and scope deny it outright. Collapsing the

first of those into a flat denial costs the operator a support ticket, which is why the state is carried through.

The whole posture is verified by 301 automated tests, including a permission matrix that asserts refusals as carefully as it asserts successes, and a secret scanner that fails the build if an RTSP URL, credential or internal hostname appears in any client-facing payload.

8.8 Performance

The edge worker reports per-stage timings as percentiles rather than averages, because the average is the summary that hides the failure. Ninety-nine frames at 40 milliseconds and one at 900 average to 48 and look healthy, while the 900 millisecond frame is the one a viewer sees as a freeze.

Measure	What it tells an operator
P50, P95 and P99 per stage	Detect, assess, enhance and OCR are timed separately, so a regression is attributable to a stage rather than to the pipeline as a whole.
Frame P99	A 25 fps feed needs this under 40 milliseconds to never fall behind. It is the number to read first.
Capacity against actual fps	Capacity is the rate this machine could sustain if it never waited. Comparing the two separates a box that is too slow from a network that delivered frames slowly.
Per-camera engine cache	One ANPR engine is kept per camera between cycles, bounded by a configured size, so a worker cycling cameras does not reload weights and discard what a camera had learned about itself.

Table 21. Performance instrumentation at the edge.

8.9 Phased rollout

Phase	Scope	What it proves
1	About 50 cameras, one district	The integration works end to end on real feeds. This is the hackathon.
2	About 2,000 cameras, 3 districts, 2 departments	Regional aggregation, adapter federation across genuinely different VMS vendors, and a real operations rhythm.
3	About 15,000 cameras, all districts, Home Department only	Statewide network load, and disaster recovery failover tested rather than designed.
4	About 80,000 cameras, all 26 departments	Steady state.

Table 22. Phased rollout. Model 1 deploys first and in full at every phase.

Model 1 deploys first and in full at every phase, because everything else needs to know what cameras exist and who owns them. It is also the phase that produces value with no analytics at all: an accurate statewide CCTV inventory with coverage-gap analysis is useful on the day it exists, and nobody currently has one.

9. Technical prerequisites and information required from departments

9.1 What we need to assess integration feasibility

Seven inputs, in the order they unblock work.

#	Input	What it determines
1	VMS make, model and version, and whether an API, SDK or ONVIF endpoint is exposed	Which integration route in section 3.3 applies, and therefore the adapter effort.
2	Camera inventory: device ID, make and model, analog or IP, resolution, codec, coordinates, view direction, purpose	Registry federation, the GIS map, coverage-gap analysis, and which analytics are worth running per camera.
3	Feed access method and reachable ports: RTSP path convention, ONVIF profile or vendor SDK	Whether live feed integration is possible at all, and over which transport.
4	Retention policy and storage location per camera group	What a playback request can return, and what the refusal should say when footage has aged out.
5	AMC status and expiry	Whether a camera can be reconfigured at all, including re-aiming one the suitability survey flags.
6	Local role vocabulary: who may view what inside the department today	The access-policy summary, and the mapping to Vigentra scope without granting local roles.
7	A technical contact who can authorise a test integration	Everything else. This is the item most likely to be the critical path.

Table 23. Information required from each participating department.

REQUEST ITEMS 3 AND 7 FIRST

Feed access and the ports reachable from the department network is the item most likely to be blocked by someone else, and it needs a named technical contact to resolve. On the sandbox grid, ports 8554 and 8889 are filtered on many networks and only 443 answers, which is why the browser only ever uses 443 and the edge worker prefers RTSP with an HLS fallback. Expect the same on department networks, and ask early. Items 1, 2, 4, 5 and 6 are paperwork and can be collected in parallel.

None of these is needed to list a department's cameras. Bulk CSV onboarding populates the registry from a spreadsheet, and item 2 alone delivers the map, the inventory and coverage-gap analysis. The rest are needed to move from registry to feed integration.

9.2 Prerequisites on the platform side

Prerequisite	Detail
Edge host per site or cluster	A machine inside the department environment that already holds the video. An accelerator only where the camera classification calls for ANPR.
Outbound metadata path	HTTPS from the edge host to the regional endpoint. No inbound connection to the camera network is required.
Central platform	Kubernetes with PostgreSQL and object storage. No licensed component, no vendor account, no API key.
Model weights	Mounted at the edge rather than baked into images, so a department can substitute weights it has validated.
Python 3.12 on the ANPR path	The consensus reader's OCR dependency publishes no wheels for newer interpreters. On a newer interpreter the worker falls back to the single-frame reader and logs one actionable line.

Table 24. Platform-side prerequisites for a department deployment.

9.3 Assumptions

- Departments retain their own VMS, storage and access policy. No migration is required or assumed.
- Camera coordinates are provided by the department or surveyed. The sandbox grid publishes none, so Vigentra's are compiled from three evidence sources with a recorded geographic confidence per camera, and a map pin never implies more precision than it has.
- Analytics run at the edge, inside the department environment that already holds the video.
- A department can accept an alert reaching a human and stopping there. Nothing in the platform issues a notice, moves a barrier or takes any automatic enforcement action.

10. Evidence: what is measured, and what is estimated

This section exists because a design document that lists only strengths is not one a deploying engineer can trust. Everything below was produced by running the system. Nothing in it is inferred from intent.

10.1 Object detection

Measured on the bundled 4K street clip, CPU only: 475 detections across six classes. Person 181, car 147, motorcycle 93, truck 26, bicycle 18, bus 10. Roughly 190 milliseconds per frame. COCO's other 74 classes are discarded rather than guessed at.

10.2 Plate detector training

Five controlled training runs on a plate-finetuned YOLO11s, deterministic with a fixed seed, each changing one variable against the run before it.

Run	What changed	mAP50	mAP50-95	P	R
A baseline 640	Deployed resolution, stock augmentation. The control.	0.650	0.446	0.877	0.556
B highres 960	Only the resolution changes.	0.630	0.416	0.798	0.578
C smallobj aug	B plus scale jitter, mosaic and small rotation.	0.602	0.391	0.729	0.533
D tiny oversample	C plus three-times exposure to tiny plates, equal step budget.	0.601	0.363	0.762	0.511
G uniform hard first	C settings, staged over real and synthetic hard cases.	0.596	0.373	0.857	0.489

Table 25. Plate detector training runs, where P is precision and R is recall. The control run is still the best of the five.

READ THIS RESULT AS A DATA CEILING, NOT A TUNING FAILURE

Higher resolution, small-object augmentation and tiny-plate oversampling each made the detector worse, and recall never cleared 0.578. Four successive attempts to improve a model by changing how it is trained, all of which regress, is evidence that the constraint is in the data distribution rather than the hyperparameters. The remedy is a plate-specific detector and a detection-free recogniser, set out in section 11.4.

10.3 The estate survey

The plate detector and the reader were run across the federated estate, 37 feed runs in total. This is the single most important measurement in this document, and it is the one most worth stating plainly.

Measure	Result
Feed runs surveyed	37
Frames processed	3,244
Vehicles tracked	3,795
Plates detected	1,961
Plates associated to a tracked vehicle	1,283
Readings produced	45
Readings confirmed by consensus	0
Median plate width, feeds carrying traffic	32 pixels
Feeds whose median plate clears the 120 pixel band	2 of 26

Table 26. Estate-wide ANPR survey across 37 feed runs.

The detector works. It found 1,961 plates and associated 1,283 of them to a tracked vehicle. The reader confirmed none of them, because the median plate on this estate is 32 pixels wide, below the 40 pixel floor at which a recogniser stops producing readings and starts producing fabrications.

That is a siting finding, not a software defect, and the survey classifies each feed accordingly: 15 feeds readable, 10 needing re-aiming, 5 marginal, and 7 carrying no traffic during the window. Of the 30 sandbox cameras, 5 are red-light violation units positioned

for number-plate work. The rest are general-purpose overview cameras mounted for situational awareness, and a camera mounted for situational awareness does not read plates, whatever software is pointed at it.

10.4 What we will not quote

The measured yield of 1 accepted plate per 67 vehicle crops, at roughly 2.5 seconds per crop on CPU, belongs to the single-frame fallback reader, not to the consensus engine. The consensus engine has not been measured on the government feed with weights installed and a camera positioned for the job, so no figure is quoted for it here. The engine's per-track voting and grammar repair are covered by automated tests; the number that matters, yield per vehicle on suitable footage, needs a run that has not been done.

Incident detection is in the same position and for the same reason. No accident footage exists for this estate, so its thresholds are reasoned rather than validated, and the only number currently knowable is how often it raises a candidate on an ordinary day.

WHY THIS SECTION IS IN THE DOCUMENT AT ALL

A competing submission can quote an ANPR accuracy figure that nobody can reproduce. This one quotes the reader it is trying not to use, names the estate survey that produced zero confirmed reads, and says what has not been measured. That is the claim we would rather be held to at deployment.

11. Limitations and the work that remains

11.1 The graded live test case, clause by clause

Clause	State	Detail
Onboard cameras	Built	30 live sandbox cameras federate across nine districts, plus two mock department systems. Manual, bulk CSV and API onboarding all work, with per-row validation and an error report.

Clause	State	Detail
Integrate live feeds	Built	The government sandbox grid is federated through the same adapter contract as everything else, over RTSP for inference with an HLS fallback where port 8554 is filtered.
Track a designated vehicle	Built	Sightings are ordered by time into a route with per-leg distance, elapsed time and implied speed. A stated reason is mandatory and audited.
Generate alerts on match	Built	Confusion-weighted matching at ingest against the active watchlist, across all five categories, with exact and near distinguished and every act audited.
Movement history and route	Built	Repeated reads at one camera collapse into one pass; impossible legs are flagged rather than dropped.
Visualise on GIS	Built	Leaflet with OpenStreetMap tiles, no API key or vendor account. Camera markers with status colouring, alert pins and a route polyline.

Table 27. The graded live test case, clause by clause.

THE HONEST QUALIFIER ON THAT TABLE

Every clause is built and works on the platform's own federated feeds. The constraint on the vehicle-tracking clause is not software: it is that most sandbox cameras are optically unsuited to reading a plate at all. Section 10.3 sets out the measurement, and section 8.2 sets out what follows from it operationally.

11.2 Limitations, stated plainly

Feature	Details
ANPR yield on wide overview footage is low, and that is optical	- Measured across 37 feed runs: zero consensus-confirmed reads, median plate width 32 pixels. See section 10.3. - A camera positioned for number-plate work reads plates. A general-purpose overview camera does not, whatever software is pointed at it.
The plate detector has plateaued on the current dataset	- Best mAP50 of 0.650 on the control run, with four successive training variations all regressing. Recall never cleared 0.578. - The remedy is architectural, not a tuning pass: a plate-specific detector in front, and a detection-free fixed-length recogniser behind it, replacing the general scene-text stack.
Facial recognition is not implemented	Section 6.5 gives the position in full: the optics do not support it here, it is a different category of decision from plate reading, and the architecture has a slot for it if a department requires it with the governance to match.
Cross-camera tracking depends entirely on plate reads	- A vehicle whose plate is never read does not appear on its own route. People are never tracked across cameras at all. - Attribute-based matching, class and colour and direction of travel within a time window at a known camera, is the designed hedge and is not yet built.
Incident thresholds are reasoned, not validated	- No accident footage exists for this estate. Every threshold should be re-derived the first time a real incident is recorded. - The module therefore emits candidates with evidence attached, never findings.
Notification is a polling console	No webhook, no server-sent events, no mobile push. Adequate for a control room with a console open, not for an officer in the field.
The metadata bus is direct HTTP, not a message queue	Defensible at this scale and named as a Model 3 feature not implemented. Section 8.4 sets out where it changes.
Coordinates are plain floats, not PostGIS	Adequate for the map rendered today, and a limitation for corridor and proximity queries. Recommended to be decided early rather than retrofitted.

Feature	Details
Edge buffering for low-connectivity sites is designed but not built	Detection IDs are deterministic and ingest is idempotent on them, so replay collapses rather than double-counting. That half is done and tested. The local queue is not.
All demonstration data is synthetic and labelled as such	- The official-source provider exists as a documented shape and refuses rather than guesses until an authorised URL and token are configured. - Vigentra does not connect to VAHAN, SARTHI, eGujCop, AFIS or NAFIS. Those integrations are designed for and gated behind the same permission and audit machinery the watchlist already uses.

11.3 Explicitly out of scope

Facial recognition, biometric identification and gait recognition. Make, model and colour verification, and vehicle re-identification by appearance. Any join between a sighting and the vehicle reference registry. Automatic enforcement of any kind: an alert reaches a human, and stops. Retroactive alerting, because matching happens at ingest rather than at query time.

11.4 Recommended next work, in order

#	Work	Why it is in this position
1	Camera suitability survey at onboarding	The estate survey already produces the verdict per feed. Making it an onboarding step is what makes the 3,000-accelerator plan real.
2	Plate-specific detector and detection-free recogniser	Removes the bottleneck behind both the 2.5 second per crop latency and the mis-association of a plate to the wrong vehicle.
3	Fine-tune on Indian plate data	Global recognisers under-weight hand-painted plates, non-standard fonts, two-line motorcycle plates and state emblems.

#	Work	Why it is in this position
4	Push notification for alerts	The polling console is the gap that most limits operational use outside a control room.
5	Attribute matching alongside plate matching	The hedge for a designated vehicle whose plate is unreadable. Far cheaper to design in than to add later.
6	PostGIS migration	A migration rather than a redesign today, and harder the longer tracking work builds on plain float columns.
7	Edge buffering queue	The idempotent half is already built and tested. Only the local queue remains.

Table 28. Recommended sequence for the work that remains.

Addendum. What changed after first issue

This proposal was first issued on 14 September 2026. That day the running system grew from 30 grid cameras to a 50-camera estate, and the wall moved from refreshed stills to live frames with detection drawn on them. Pages 16, 17, 40 and 43 are revised to match.

A.1 The fifty-camera estate

Source	Cameras	How it reaches Vigentra
Sentinel government grid	30	Grid adapter, over RTSP, with credentials held server-side
Traffic Police VMS, own Delhi test clip	1	Installation request, served by the VMS
Traffic Police VMS, TfL JamCam	19	Installation requests; public clips relayed by the VMS. Powered by TfL Open Data

Table A1. The fifty cameras in service, by source.

All 50 are in service and online. The last 20 were onboarded through the Traffic Police VMS installation-request flow. The registry holds 70 cameras: 50 in service and 20 decommissioned, kept for the audit trail.

A.2 Live frames with live detection on the wall

An edge frame service decodes every feed in a bounded pool and keeps the newest frame, and a background annotator draws YOLO11n vehicle detection on it. A tile is answered from memory in about 2 ms. VMS feeds open through the VMS's own authorised live handle. A tile holds a session only while on screen; click it for full screen, and Esc returns.

A.3 Live ANPR on chosen feeds

The full Vigentra ANPR engine runs live on chosen feeds: the Delhi test clip and grid CAM06, Madhuram Bypass Road, Bhavnagar. Each read is drawn with its confidence, and

a low one is shown as low. The engines share one lock, as the GPU backend is not safe across threads, and each live reader releases its own capture. That removed two crashes found under load.

A.4 Keeping the API responsive under the wall

The snapshot route checks permission in a short database session, releases it, then fetches the frame with a pooled HTTP client. Under the full wall, API health now answers in 0.02 to 0.1 seconds, where it took 5 to 15. The open-alert badge every page polls is a count and is deliberately not audited; reading the alerts themselves still is.

A.5 Console fixes found while recording the demonstration

- Page filters and inputs survive a reload, per tab, and never a password.
- Enter submits the sign-in and search forms.
- The registry opens on in-service cameras and fits the map to Gujarat.
- The logo's tagline is set as text, so Safari lays it out correctly.

A.6 Measured, 15 September 2026

Measure	Value
Cameras in service	50 of 50, all online
Detections	278,362 vehicles and objects, cumulative
Plate readings	870, across 15 cameras
Automated tests	301 passing (196 platform, 105 edge worker), 4 skipped
API health, full wall	0.02 to 0.1 s, was 5 to 15 s

Table A2. Measured on 15 September 2026. Skipped tests need video fixtures kept out of the repository.

Pages revised in this issue: 16 (transport fallback), 17 (the live wall), 40 (test count) and 43 (ports and the browser).

Addendum B. Reading plates right, and alerts that arrive

Between 17 and 20 September 2026 the plate reader was measured against ground truth read by eye on every recorded clip, and the alert path was brought in line with the challenge's requirement for real-time, prioritised alerts. Nothing earlier in this proposal is withdrawn.

B.1 ANPR, measured against what a person can read

Measure, over the recorded clips	Before	Now
Plates read correctly, 36-clip set (26 legible by eye)	9	16
All 38 clips, with grid CAM06 at noon and at night (48 legible)	not run	36
Grid CAM06 at noon: 2 minutes, 84 vehicles, 16 legible	not run	15
False plates confirmed or sent, any clip	0	0
Vehicle tracks, 36-clip set	1,524	1,868

Table B1. Whole clips, Apple M1. Ground truth is by eye and is used only for scoring.

The gains came from decision rules, not a new model: one vote per vehicle weighted by crop quality, detection at each camera's own resolution, a per-camera configuration file, and confirmation when a character reader and a whole-plate text reader agree. The grid's 720p night feeds carry plates 14 to 31 px wide; nobody can read them, and the system reads none rather than guess.

B.2 From a read to an alert, in seconds

- A plate is uploaded with the frame that settles it, no longer with the batch or at the end of a pass; the alert console refreshes every 5 seconds.
- Every alert carries a priority: critical for an exact read of a stolen or wanted vehicle, high for any other exact read, review for a near match. Open alerts lead the console, most urgent first.
- Each alert is also sent to a webhook, signed with HMAC-SHA256, so dispatch, SMS or messaging can sit behind it. A receiver that is down never delays an ingest.
- Raise, acknowledge and dismiss are each audited; a dismissal needs a reason.

B.3 Continuous readers for the designated-vehicle test

The sampled worker visits each camera for about 20 seconds and returns minutes later. That counts traffic well and does not see one vehicle cross a junction. Cameras whose plates are legible now get a continuous reader: long passes, a 3-second breath between them, and the full pause only after a failure, so a dead feed is never hammered. One 720p camera runs at 1.00 times real time on one M1-class machine, which is the sizing unit for the finale.

B.4 Intrusion, and the position on faces

A camera's profile may carry a restricted zone: a polygon, a dwell and the hours it is armed. A person whose foot point stays inside raises an intrusion incident. A face matcher would plug in behind the same watchlist and alert path; it is not built, because this footage carries a face no better than a 14 px plate.

B.5 Measured, 20 September 2026

Measure	Value
Plate ingest with watchlist matching	164 reads a second through one API process, 200 watched plates
ANPR at camera speed	720p 1.00x, 854x480 1.31x, 1080p 0.92x real time, one machine

Table B2. One Apple M1 laptop, 20 September 2026. 332 automated tests pass (201 platform, 131 edge worker).